GENETICS CROSS PROBLEMS

Step-by-Step Student Handout

Outcome 30-C2.2k — Genotype & Phenotype Ratios and Probabilities

KEY VOCABULARY

Gene: A segment of DNA that codes for a specific trait.	**Allele:** A variant form of a gene. Organisms carry two alleles per gene.
Genotype: The genetic makeup — the alleles an organism carries (e.g., Aa, BB).	**Phenotype:** The observable physical expression of the genotype (e.g., tall, Type A blood).
Dominant: An allele expressed even with only one copy. Written UPPERCASE (A).	**Recessive:** An allele expressed only when two copies are present. Written lowercase (a).
Homozygous: Two identical alleles for a gene (AA or aa).	**Heterozygous:** Two different alleles for a gene (Aa).
Punnett Square: A diagram used to predict possible offspring genotypes and phenotypes.	**Polygenic:** Traits controlled by two or more genes producing a continuous range of phenotypes.

TYPES OF ALLELE INTERACTIONS

Type	Heterozygous Phenotype	Notation	Example
Dominant/Recessive	Only dominant trait visible	A (dominant), a (recessive)	Tall (T_) vs. Short (tt)
Incomplete Dominance	Intermediate BLEND of both traits	R^1R^2	Red x White = Pink flowers
Codominance	BOTH traits fully visible simultaneously	R^BR^W	Roan cattle (black + white hairs)
Multiple Alleles	Depends on specific allele combo	I^A, I^B, i	ABO blood types (A, B, AB, O)
Polygenic	Continuous range / spectrum	AaBb, AaBbCc...	Height, skin colour, hair colour

MONOHYBRID CROSS — Dominant & Recessive Alleles

A **monohybrid cross** tracks inheritance of a single gene with two alleles (one dominant, one recessive). Use a 2×2 Punnett Square.

1	**Identify the trait and alleles:** Decide which allele is dominant (capital) and which is recessive (lowercase).
2	**Write parental genotypes:** Record the genotype of each parent (e.g., Tt and Tt).
3	**Determine the gametes:** Each parent passes ONE allele. Separate alleles: Tt → T and t.
4	**Draw the Punnett Square:** Place one parent's gametes along the top; the other's down the left side.

5	**Fill in the boxes:** Each box = top allele + side allele. Write capitals first (Tt not tT).
6	**Count and interpret results:** Tally each genotype. Assign phenotypes. Write as ratios and probabilities.

Worked Example 1: Cross two heterozygous tall pea plants (Tt × Tt). Predict offspring genotypes and phenotypes.

	T	t
T	**TT**	**Tt**
t	**Tt**	**tt**

Genotype Results:

1/4	2/4	1/4
TT (25%)	Tt (50%)	tt (25%)

Phenotype Results:

3/4	1/4
Tall (75%)	Short (25%)

Monohybrid Ratios (Tt × Tt):
Genotype: 1 TT : 2 Tt : 1 tt (1:2:1)
Phenotype: 3 Tall : 1 Short (3:1)

Probability Summary — Tt × Tt Cross

Genotype	Fraction	%	Phenotype	Phenotype Probability
TT — Homozygous dominant	1/4	25%	Tall	
Tt — Heterozygous	2/4	50%	Tall	
tt — Homozygous recessive	1/4	25%	Short	**1/4 = 25% Short**

INCOMPLETE DOMINANCE

Neither allele is fully dominant. The heterozygote shows an **intermediate blend** of the two homozygous phenotypes. Notation: use superscripts (R^1, R^2) since neither allele dominates.

Worked Example — Snapdragon Flower Colour:
F1 Cross: R^1R^1 (Red) × R^2R^2 (White) → All R^1R^2 = PINK (intermediate blend)
F1 × F1 Cross: R^1R^2 × R^1R^2

	R^1	R^2
R^1	R^1R^1	R^1R^2
R^2	R^1R^2	R^2R^2

F1 × F1 Results:

1/4	2/4	1/4
R^1R^1 Red	R^1R^2 Pink	R^2R^2 White

Phenotype Ratio: 1 Red : 2 Pink : 1 White
Note: Genotype ratio = Phenotype ratio (1:2:1)

CODOMINANCE

Both alleles are **fully expressed simultaneously** in the heterozygote. Both phenotypes appear side-by-side — NOT a blend. Notation: same as incomplete dominance (superscripts).

Worked Example — Roan Cattle:
Cross two roan cattle: $R^BR^W \times R^BR^W$
Roan = individual shows BOTH red AND white hairs (not a pink blend)

	R■	R■
R■	R■R■	R■R■
R■	R■R■	R■R■

Results:

1/4	2/4	1/4
R■R■ Black	R■R■ Roan	R■R■ White

Phenotype Ratio: 1 Black : 2 Roan : 1 White
Roan cattle show both black AND white hairs — not pink!

MULTIPLE ALLELES — ABO Blood Types

Some genes have **more than two allele forms** in the population. Each individual still carries only two alleles. The ABO blood type system has three alleles: I^A, I^B, and i.

Allele	Dominance Relationship	Blood Antigen
I^A	Dominant over i; codominant with I^B	A antigen on red blood cells
I^B	Dominant over i; codominant with I^A	B antigen on red blood cells
i	Recessive — only expressed when homozygous (ii)	No antigen (Type O)

Genotype	Blood Type	Antigen(s)	Notes
I^AI^A	Type A	A antigen	Homozygous dominant
I^Ai	Type A	A antigen	Heterozygous; i is recessive
I^BI^B	Type B	B antigen	Homozygous dominant
I^Bi	Type B	B antigen	Heterozygous; i is recessive

$I^A I^B$	Type AB	Both A and B	Codominance — both expressed
ii	Type O	Neither	Homozygous recessive

Worked Example — ABO Cross:
Parent 1: Type A heterozygous ($I^A i$) × Parent 2: Type B heterozygous ($I^B i$)
Offspring: 1/4 Type AB ($I^A I^B$), 1/4 Type B ($I^B i$), 1/4 Type A ($I^A i$), 1/4 Type O (ii) — all four blood types possible!

DIHYBRID CROSS — Two Genes Simultaneously

A **dihybrid cross** tracks TWO separate genes at the same time using a 4×4 Punnett Square. Genes on different chromosomes are inherited independently (Mendel's Law of Independent Assortment).

1	**Identify both traits and alleles:** E.g., Seed shape: R = round (dom), r = wrinkled (rec); Seed colour: Y = yellow (dom), y = green (rec).
2	**Write parental genotypes:** E.g., Parent 1 = RrYy × Parent 2 = RrYy (heterozygous for both).
3	**Determine the 4 gamete types:** Each gamete carries one allele from each gene. For RrYy: gametes are RY, Ry, rY, ry.
4	**Draw a 4×4 Punnett Square:** Place 4 gametes from one parent along the top; 4 from the other parent down the side.
5	**Fill in all 16 boxes:** Each box = top gamete + side gamete. Write capitals within each gene first.
6	**Count phenotypes and calculate ratios:** Classify each box by phenotype. Expected ratio for two heterozygotes: 9:3:3:1.

Worked Example — Pea Seed Shape and Colour (RrYy × RrYy):
Gametes from each parent: RY | Ry | rY | ry

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Round Yellow (R_Y_) — 9/16 = 56.25%
Round Green (R_yy) — 3/16 = 18.75%
Wrinkled Yellow (rrY_) — 3/16 = 18.75%
Wrinkled Green (rryy) — 1/16 = 6.25%

Dihybrid Phenotype Ratio: 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green
Shortcut: Multiply individual probabilities — P(Round, Yellow) = 3/4 × 3/4 = 9/16

POLYGENIC TRAITS — Continuous Variation

Polygenic traits are controlled by **two or more genes**, each contributing additively to the phenotype. Instead of distinct categories, polygenic traits produce a **continuous range (spectrum)** of phenotypes in a population, forming a bell-shaped distribution.

Key Features of Polygenic Traits:
• Controlled by multiple gene loci — each gene contributes a small, additive effect
• More dominant alleles = more of the trait expressed
• Distribution in a large population forms a bell-shaped (normal) curve
• Often also influenced by environmental factors (nutrition, temperature, etc.)
• Cannot be neatly separated into 2 or 3 phenotype classes

Example 1: Skin Colour (Simplified 2-Gene Model)

Assume two genes (A/a and B/b) control pigmentation. Each dominant allele adds one "unit" of pigmentation. Cross AaBb × AaBb (two medium-toned individuals):

# of Dominant Alleles	Genotype Examples	Skin Tone	Boxes (out of 16)	Fraction
4 (darkest)	AABB	Very dark	1	1/16
3	AABb, AaBB	Dark	4	4/16
2 (medium)	AAbb, AaBb, aaBB	Medium	6	6/16
1	Aabb, aaBb	Light	4	4/16
0 (lightest)	aabb	Very light	1	1/16

Genotypic / Phenotypic Ratio: 1 : 4 : 6 : 4 : 1 — This mirrors a binomial expansion. With more genes involved, more phenotype classes appear, approaching a smooth bell curve.

Example 2: Human Height (Multi-Gene Polygenic Trait)

Height is influenced by **hundreds of genes** plus environmental factors (nutrition, health). Each gene contributes a small additive effect. The simplified 3-gene model below illustrates the bell-curve pattern:

"Tall" Alleles	Fraction (3 genes)	Height Category	Approx. Height
0 (aabbcc)	1/64	Very short	~150 cm
1	6/64	Short	~155 cm
2	15/64	Below average	~160 cm
3 (average)	20/64	Average	~168 cm
4	15/64	Above average	~175 cm
5	6/64	Tall	~183 cm
6 (AABBCC)	1/64	Very tall	~190+ cm

Comparing Mendelian vs. Polygenic Traits

Feature	Simple Mendelian	Polygenic

Number of genes	1	2 or more
Phenotype categories	2–3 distinct classes	Continuous range / spectrum
Population distribution	Distinct ratios (3:1, 1:2:1)	Bell-shaped (normal) curve
Environmental influence	Little to none	Often significant
Examples	Pea colour, blood type, tongue rolling	Height, skin colour, hair colour, weight

SUMMARY — Comparing All Allele Types (30-C2.2k)

Allele Type	Heterozygous Phenotype	Genotype Ratio (Aa × Aa)	Phenotype Ratio
Dominant/Recessive	Dominant trait only	1 AA : 2 Aa : 1 aa	**3 dom : 1 rec**
Incomplete Dominance	Intermediate blend	1 : 2 : 1	**1 : 2 : 1 (3 phenotypes)**
Codominance	Both traits fully expressed	1 : 2 : 1	**1 : 2 : 1 (3 phenotypes)**
Multiple Alleles (ABO)	Depends on combination	6 possible genotypes	**Up to 4 phenotypes**
Dihybrid Cross	Both dominant traits shown	9 genotype classes	**9 : 3 : 3 : 1**
Polygenic	Continuous spectrum	1:4:6:4:1 (2 genes) More genes → bell curve	**Continuous distribution**

Key Rule: For dominant/recessive crosses, genotype ratio (1:2:1) ≠ phenotype ratio (3:1) because TT and Tt look identical. For incomplete dominance and codominance, genotype = phenotype ratio (1:2:1) because each genotype produces a uniquely identifiable phenotype.

PRACTICE PROBLEMS

1

Monohybrid — Dominant/Recessive
In guinea pigs, black coat (B) is dominant over white coat (b). Cross a heterozygous black guinea pig (Bb) with a white guinea pig (bb). What are the expected genotype and phenotype ratios of the offspring?

Work space:

2

Incomplete Dominance
In four-o'clock plants, red flowers (R^1R^1) crossed with white flowers (R^2R^2) produce all pink (R^1R^2) offspring. If two pink plants are crossed, what phenotypes and in what ratio would you expect?

Work space:

3 Codominance

A black rooster (R■R■) is crossed with a white hen (R■R■). What is the phenotype of all F1 offspring? If two F1 birds are crossed, what are the expected offspring phenotypes and ratios?

Work space:

4 Multiple Alleles

A woman with Type AB blood (I■I■) has children with a man who has Type O blood (ii). List all possible blood types their children could have and the probability of each.

Work space:

5 Dihybrid Cross

In peas, round seed (R) is dominant over wrinkled (r), and yellow seed (Y) is dominant over green (y). Cross RrYy × Rryy. Determine the expected phenotype ratios for the offspring. (Hint: use a 4×2 Punnett Square or multiply probabilities.)

Work space:

6 Polygenic Traits

Explain why you cannot use a simple Punnett Square to predict the exact height of a child whose parents are both of average height. What pattern would you expect in the heights of many children from this population? Include the terms "polygenic," "additive," and "bell-shaped distribution" in your answer.

Work space: